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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CO1) | Unit 2, Computing Career Pathways

Explore Career Pathways

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Course/Program: BSCS	**Date:** September 16, 2026

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just "programming." As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science

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& AI Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

6. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you don't.

Part E – Real-World Job Search (Optional / Bonus)

7. Search a real job platform (e.g., LinkedIn, Indeed, JobStreet, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	Designing and building application and algorithm driven software system	Software Engineer, Full Stack Engineer
Systems & Cloud Infrastructure	Maintaining hardware provisions and deployment pipelines	Systems Administrator, Site Reliability Engineer (SRE)
Data Science & AI Engineering	Extracting insights from structured and unstructured data.	Data Scientist, Data Engineer
Cybersecurity & Threat Defense	Safeguarding network perimeter, and managing incident response.	Information Security Analyst, SOC Specialist
Information Systems & Enterprise IT	Aligning business workflows with technology solutions and enterprise software architectures.	Business System Analyst, IT Manager

My Chosen Pathway (Part B)

I am most interested in:

Why I chose it:

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Software Engineering, As a CS student, I am interested in software engineering because I enjoy problem-solving and turning ideas into real applications. It excites me to build software that can help people and make tasks easier. This job also allows me to continuously learn and improve my coding skills.

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Entry-Level	Junior Type Coder	Coding And Debugging, Creating Frameworks
Mid-Level	Mid-Level type Coder	Unit Testing, Algorithm Efficiency
Senior-Level	Senior Level Engineer	System Architecture Design, Scalability and Performance
Expert / Leadership	Technical Leader Engineeri	Architectural Overview, Cross-functional Collaboration

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Coding and Debugging	Y	1. I will further enhance this skill by creating personal projects and participating in coding exercises.
Creating Frameworks	N	2. I plan to learn it by taking online courses and watching tutorials about modern frameworks.
Unit Testing	N	3. I will learn it through our Software Engineering lessons and by applying it to my future system projects.
Algorithm Efficiency	Y	4. I will improve it by practicing data structures and algorithms regularly on online platforms.
System Architecture Design	N	5. I intend to learn it by studying system design concepts and analyzing existing software architectures.

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Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Scalability and Performance	N	6. I plan to learn it through research, self-study, and by seeking guidance from professors and peers.

Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title	Junior Software Developer
Company	Accenture Philippines
Platform / Source (e.g., LinkedIn, Indeed, JobStreet)	LinkedIn
Required Skills Listed (3–5)	Knowledge in Java, Python, or C++ Basic understanding of Software Development Life Cycle (SDLC) Good problem-solving and analytical skills Ability to work in a team and good communication skills

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

I chose this pathway because I have always been curious about how computers and applications work behind the screen. I enjoy logic and coding, and it feels rewarding when I fix a bug and see my program run. I also chose it because I see many opportunities in this field to grow and help others through technology through software.

2. Looking at your roadmap, which career stage excites you the most, and why?

Maybe my Entry level because it just the start but enjoying at the same time, so it excites me specially it is my chosen pathway.

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

The skill I already have is basic problem-solving and familiarity with programming logic from our CS subjects. The hardest for me to develop will be advanced software design and working with large, complex systems. It will take more time, practice, and real project experience to master that competency.

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

Mapping this roadmap confirms that I am on the right track and motivates me to stay focused for the rest of the course. It helps me see which skills I need to improve, like programming and problem-solving, before I graduate. It also changes my plan by making me more serious about building projects and learning beyond the classroom.

VII. Scoring Guide

Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date

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